

Improving Persian News Headline Generation using LoRA Fine-tuning on Llama 3.1

Farzad Jannati
farzadjannati@ut.ac.ir

Shahriar Rahimi Rad
rahimirad@ut.ac.ir

Abolfazl Assaraian Nejad
abolfazl.assarian@ut.ac.ir

1 Problem statement

The generation of compelling and factually accurate news headlines is a critical yet challenging task in abstractive text summarization. While Large Language Models (LLMs) have demonstrated remarkable fluency, their application to specialized domains, particularly for morphologically rich and lower-resource languages like Persian, presents significant hurdles. General-purpose models often fail to capture the nuanced context of Persian news, leading to headlines that, while fluent, may suffer from factual inaccuracies or “hallucinations.” This issue is particularly detrimental in journalism, where accuracy and reader trust are paramount (10).

The core problem this research addresses is the lack of a specialized model adept at generating high-quality Persian headlines that are both engaging and, most importantly, factually consistent with the source article. While standard Supervised Fine-Tuning (SFT) can adapt models to a specific domain, it does not explicitly penalize factual divergence. Recent advancements, such as the methodology proposed in CAP-LLM (8), have introduced multi-task learning with contrastive objectives to enforce factual alignment for English. However, a significant research gap exists in adapting and evaluating these state-of-the-art, fact-aware fine-tuning techniques for the Persian language.

This project aims to bridge this gap by developing and rigorously evaluating a model for Persian news headline generation that prioritizes factual consistency. Our motivation stems from the need for a reliable automated journalism tool that can support Persian news agencies by producing high-quality,

trustworthy headlines. We achieve this by fine-tuning the state-of-the-art Llama 3.1 8B Instruct model using the parameter-efficient LoRA technique, enhanced with a novel multi-task learning objective that incorporates a contrastive loss to explicitly minimize factual errors.

Our implementation is available as a [public GitHub repository](#).

2 What you proposed vs. what you accomplished

Our project successfully implemented the core objectives outlined in the initial proposal and significantly expanded upon them by incorporating a more advanced, state-of-the-art methodology. Below is a summary of our progress:

- ~~Collect and preprocess the pn-summary dataset.~~ We successfully processed this large-scale Persian news dataset, including comprehensive cleaning and preparation for fine-tuning.
- ~~Build and train baseline models on collected data.~~ We established a robust evaluation framework by running inference on four strong baseline models: Gemma 3 4B, Qwen 3 8B, Mistral 7B, and the base Llama 3 8B model. Results were benchmarked using ROUGE and BERTScore.
- ~~Build and train a “fancy model” by fine-tuning Llama 3 with LoRA.~~ We not only accomplished this but exceeded our initial plan. Instead of only a standard Supervised Fine-Tuning (SFT), we successfully implemented an advanced multi-

task learning approach inspired by the CAP-LLM paper. This involved a custom trainer that integrates a contrastive loss function to explicitly enhance the factual consistency of the generated headlines, directly addressing a core challenge of the problem.

- *Specialize the model for sensational headlines using sports news.* While we proposed this as a potential subsequent phase, we focused our efforts on mastering the more fundamental challenge of factual consistency. We have instead defined this stylistic adaptation as a clear direction for future work, aiming to implement a Click-based Personalization and Training (CPT) system.
- ~~Perform in-depth error analysis and evaluation.~~ We conducted a comprehensive evaluation using both automated metrics (ROUGE, BERTScore) and qualitative analysis to assess the performance of our models, identifying key areas of success and opportunities for future improvement.

3 Related Work

The field of automatic news headline generation, a specialized subfield of abstractive text summarization, has evolved significantly with the advent of Large Language Models (LLMs) (1). Early approaches transitioned from statistical methods to sequence-to-sequence architectures, with models like BART setting strong benchmarks by pre-training on large-scale corpora (2). However, the emergence of powerful foundation models like the Llama series has shifted the paradigm towards leveraging their immense generative capabilities through fine-tuning, as demonstrated in recent benchmarking studies (3). While these models excel at producing fluent text, their application in high-stakes domains like journalism necessitates addressing critical challenges, namely factual consistency and user engagement (10).

A primary challenge in generative models is the mitigation of “hallucinations”—the generation of factually incorrect or unsupported information. This is particularly detrimental in

news reporting, where trust and accuracy are paramount. To address this, a significant line of research has focused on enforcing factual alignment. The pioneering CAP-LLM framework by Wilson et al. (8) introduced a novel approach by incorporating a *contrastive loss* function during fine-tuning. This method explicitly penalizes factual divergence between the generated headline and the source article, effectively steering the model towards greater accuracy. Our work is heavily inspired by this methodology. While CAP-LLM focuses on English news, we adapt its core principle of contrastive learning to the unique challenges of the Persian language. The importance of aligning model outputs with human judgments of quality and accuracy is further underscored by research exploring advanced techniques like Reinforcement Learning from Human Feedback (RLHF) (11).

Concurrently, another research thrust has concentrated on the pragmatic quality of headlines, moving beyond simple informativeness to optimize for user engagement and personalization. The goal is to create headlines that are not only accurate but also “catchy” and tailored to individual user interests. Initial efforts focused on creating datasets and models for controllable and interpretable generation (6). More advanced frameworks like LOLA (7) and SCAPE (5) have explored generating personalized headlines by modeling user click behavior and stylistic preferences. The work on “Panoramic Interests” further argues that true personalization must consider not just a user’s content preferences but also their stylistic inclinations, such as a preference for interrogative or list-based headlines (4). Our future work, which aims to implement a Click-based Personalization Training (CPT) system, directly builds upon the concepts presented in these papers.

Our work is positioned at the confluence of these research streams. While the aforementioned studies have made significant strides for high-resource languages like English, and some adaptations exist for other languages like Thai (9), the application of these state-of-the-art techniques to morphologically rich and lower-resource languages like Persian remains largely underexplored. Our key con-

tribution is the adaptation and fine-tuning of the state-of-the-art Llama 3.1 model using the efficient LoRA technique, specifically tailored for Persian news. Differentiating from standard supervised fine-tuning, our approach, inspired by CAP-LLM (8), integrates a contrastive loss into the training objective to explicitly enhance the factual consistency of generated headlines. By doing so, we aim to establish a robust and fact-aware baseline for Persian news headline generation, bridging the gap between advanced generative techniques and the specific linguistic needs of this domain.

4 Your dataset

Our project addresses the task of abstractive headline generation for Persian news, leveraging the comprehensive PN-Summary dataset, sourced from the HooshvareLab repository on Hugging Face. This dataset provides a robust foundation for our fine-tuning experiments, consisting of 82,022 samples in the training set and 5,593 in the test set. Each data point contains a full news article, a human-written summary, and a corresponding headline, collected from prominent Iranian news agencies.

The primary challenge of this task lies in the inherent nature of the Persian language—its rich morphology and syntactic flexibility—combined with the journalistic requirement for headlines to be not only factually consistent with the source text but also concise and engaging. A simple extractive summary is insufficient; the model must learn to synthesize information and generate novel, stylistically appropriate text.

To deeply understand the dataset’s characteristics, we performed a thorough statistical analysis, the results of which are visualized in Figure 1. Our analysis revealed that headlines are significantly more concise than summaries, with an average length of approximately 9 words (50 characters), compared to summaries which average around 29 words (154 characters). This notable length discrepancy underscores the abstractive nature of the task. Furthermore, the dataset spans a wide range of categories, with topics like Oil & Energy, Local, and Economy being the most prevalent. This diversity is crucial for training a model that can generalize across different news do-

main.

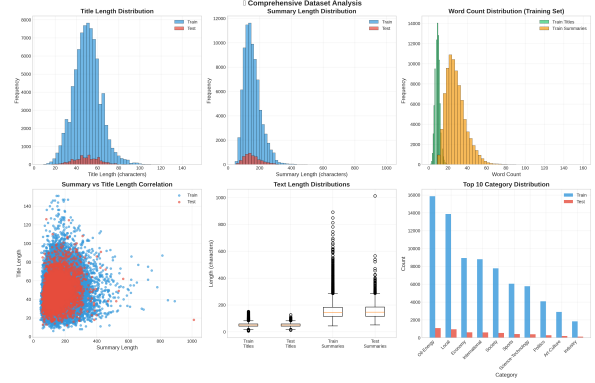


Figure 1: A comprehensive statistical analysis of the PN-Summary dataset. The plots illustrate (top row) the distribution of title lengths, summary lengths, and word counts; and (bottom row) the correlation between summary and title length, box plots of text length distributions, and the distribution of the top 10 news categories.

For clarity, a typical input-output pair from our preprocessed dataset is as follows. The model is tasked with generating the headline based on the provided summary:

Input (Summary):

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Output (Ground Truth Headline):

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4.1 Data preprocessing

To ensure the model receives high-quality and consistent input, we implemented a multi-stage preprocessing pipeline, a critical step detailed in our project workflow. This pipeline was applied to both the title and summary fields of the dataset. The key steps were as follows:

- **Normalization:** We utilized the `hazm` library, a standard toolkit for Persian NLP, to perform initial text normalization. This step standardizes character representations (e.g., unifying different variations of spaces and punctuation) and prepares the text for further processing.
- **Character Correction:** A crucial step for Persian text is the conversion of Arabic characters to their Persian equivalents (e.g., ' ' ' ' ' ' .(' ' This ensures a consistent character set, which is vital for the

model’s vocabulary and embedding layers.

- **Noise and Boilerplate Removal:** News articles often contain irrelevant boilerplate text. We developed a set of regular expressions to systematically remove common noise patterns such as reporter attributions (e.g., ”,(“.. structural markers like ”[n]”, and embedded URLs. This cleaning process helps the model focus on the core semantic content of the news summary.
- **Spacing Correction:** We applied further regex-based corrections to rectify common spacing errors, such as inconsistent spacing around punctuation, which improves tokenization and model comprehension.
- **Quality Filtering:** After the initial cleaning, a final quality filtering step was performed. We calculated the word count for all cleaned titles and summaries and removed samples where the title was shorter than 3 words or the summary was shorter than 10 words. This heuristic helps eliminate erroneous or low-quality data points, such as entries with empty fields or placeholder text. This filtering step removed a negligible portion of the data—only 97 training samples (0.12%) and 8 test samples (0.14%)—thereby improving dataset quality without significant data loss.

4.2 Data annotation

This project did not require a manual data annotation phase as we utilized the pre-existing, high-quality PN-Summary dataset. This dataset, with its large volume of professionally written article-headline pairs, provided a robust foundation for our fine-tuning experiments. This allowed us to focus our efforts on the architectural and methodological improvements of the model, such as implementing the contrastive loss function for factual consistency, rather than on data creation.

5 Baselines

To establish a robust performance benchmark for our proposed approach, we evaluated a set

of four powerful, state-of-the-art open-weight Language Models (LLMs) in a zero-shot inference setting. The selected baseline models represent a diverse range of architectures and developers, ensuring a comprehensive comparison. These models are:

- **Gemma 3 4B-IT** (google/gemma-3-4b-it): A highly capable instruction-tuned model from Google.
- **Qwen 3 8B** (Qwen/Qwen3-8B): A powerful multilingual model developed by the Qwen team.
- **Mistral 7B Instruct v0.3** (mistralai/Mistral-7B-Instruct-v0.3): A widely recognized and high-performing model from Mistral AI.
- **Llama 3.1 8B Instruct** (meta-llama/Llama-3.1-8B-Instruct): The latest instruction-tuned model from Meta AI, serving as the foundation for our fine-tuning work.

These models were chosen for their strong performance on general NLP tasks and their public availability, making them suitable candidates for establishing a challenging and reproducible baseline. No fine-tuning or hyperparameter optimization was performed on these models; our objective was to assess their out-of-the-box capabilities for Persian news headline generation when guided by a carefully engineered prompt. The evaluation was conducted on the first 100 samples of the test split from our preprocessed `pn_summary` dataset. Each model was loaded with `torch.float16` precision and prompted to act as a professional journalist, tasked with writing an “attractive, short and catchy headline in Persian” for the given news text.

The performance of each baseline model was quantitatively assessed using two primary families of automated metrics: ROUGE (for lexical overlap) and BERTScore (for semantic similarity). The aggregated results are presented in Table 1.

A key observation from the results is the striking disparity between ROUGE and BERTScore metrics. The ROUGE scores for

Table 1: Automatic evaluation metrics for baseline models on the first 100 test samples. Note the discrepancy between the near-zero ROUGE scores and the higher, more meaningful BERTScore F1 values.

Model	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore F1
Gemma-2-4B	0.0000	0.0000	0.0000	0.6586
Qwen2.5-3B	0.0000	0.0000	0.0000	0.6535
Mistral-7B	0.0000	0.0000	0.0000	0.6740
LLaMA-3.1-8B	0.0000	0.0000	0.0000	0.6680

all baseline models are effectively zero. This indicates a significant lexical divergence between the generated headlines and the ground-truth references. Manual inspection of the outputs confirmed that the zero-shot models often produced verbose responses, sometimes including conversational filler or rephrasing the prompt, which resulted in a failure to match the n-grams of the concise reference headlines.

In contrast, the BERTScore F1 results are substantially higher (ranging from 0.65 to 0.67), with Mistral 7B achieving the highest score. This suggests that while the models struggled with generating lexically identical headlines, they were capable of producing text that was semantically relevant to the source article. This discrepancy highlights a known limitation of ROUGE for evaluating abstractive summarization and underscores the importance of using semantic metrics like BERTScore for a more nuanced assessment.

The visual comparisons in Figure 2, Figure 3, and the summary in Figure 4 further illustrate these findings. The ROUGE score chart confirms the poor lexical performance across all models, while the BERTScore chart provides a clearer picture of their relative semantic capabilities. The performance heatmap consolidates these metrics, visually representing Mistral 7B’s slight edge in semantic relevance. Overall, these results demonstrate that while powerful, general-purpose LLMs possess a foundational understanding of the task, they require specialized fine-tuning to adhere to the specific stylistic and conciseness constraints of high-quality headline generation.

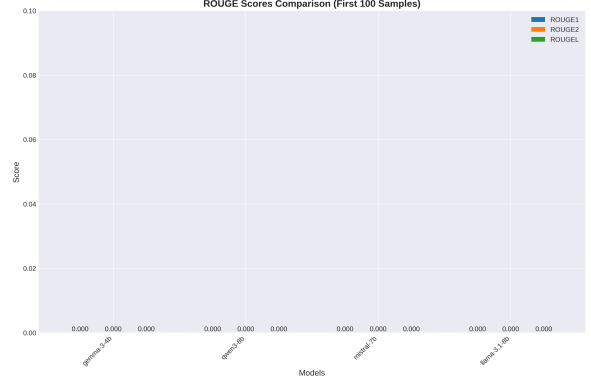


Figure 2: ROUGE Scores Comparison for Baseline Models. The chart visually confirms the near-zero scores for ROUGE-1, ROUGE-2, and ROUGE-L, indicating a significant lack of lexical overlap with reference headlines in the zero-shot setting.

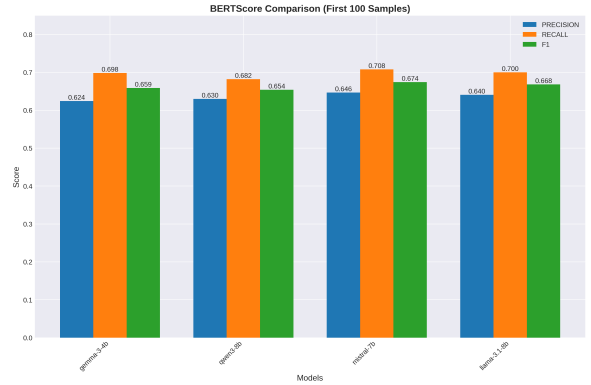


Figure 3: BERTScore Comparison for Baseline Models. Unlike ROUGE, BERTScore reveals a stronger semantic performance, with Mistral 7B slightly outperforming the other models in F1 score.

6 Your approach

Our approach moves beyond zero-shot prompting to specialize the state-of-the-art `meta-llama/Llama-3.1-8B-Instruct` model for the nuanced task of Persian news headline generation. The core of our methodology is Parameter-Efficient Fine-Tuning (PEFT) using Low-Rank Adaptation (LoRA), which allows for the efficient adaptation of this large model on standard high-performance computing instances.

We implemented and rigorously compared two distinct fine-tuning strategies. The first is a standard Supervised Fine-Tuning (SFT) approach, which serves as a strong internal baseline. The second, which represents our primary contribution, is an advanced multi-task

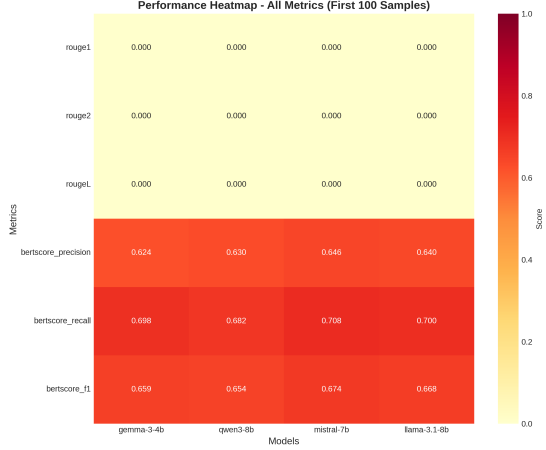


Figure 4: Performance Heatmap of All Metrics for Baseline Models. This visualization provides a consolidated view of the evaluation scores, highlighting the models’ relative strengths in semantic similarity (BERTScore) versus their universal weakness in lexical overlap (ROUGE).

learning framework inspired by the CAP-LLM methodology (8). This approach is explicitly designed to enhance the factual consistency of the generated headlines, a critical requirement in the journalistic domain.

6.1 Technical Architecture and Multi-task Objective

The technical architecture of our proposed model, shown in Figure 5, enhances the base Llama 3.1 model with LoRA adapters and a composite loss function designed to balance fluency and factual accuracy.

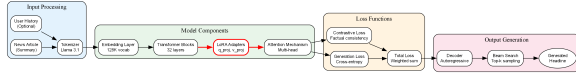


Figure 5: The technical architecture of our proposed model. The input summary is processed by the Llama 3.1 model with LoRA adapters. The model’s outputs are used to compute a multi-task loss, combining a standard generation loss with a contrastive loss for factual consistency.

Parameter-Efficient Fine-Tuning (PEFT): To make fine-tuning an 8-billion-parameter model computationally feasible, we employed LoRA. This technique freezes the pre-trained model weights and injects trainable, low-rank matrices into the Transformer architecture’s attention mechanism. As specified in our implementation (`PersianTitleGenerator.json`), we applied

LoRA to the query (`q_proj`) and value (`v_proj`) projection matrices with a rank (`r`) of 16 and an alpha of 32. This drastically reduced the number of trainable parameters to approximately 6.8 million (0.08% of the total), enabling efficient training on a single GPU.

Multi-task Learning Objective: Our primary contribution is the implementation of a multi-task loss function to improve factual consistency. The total loss is a weighted sum of a standard generation loss and a factual consistency contrastive loss:

$$\mathcal{L}_{total} = \mathcal{L}_{gen} + \lambda_{fact}\mathcal{L}_{fact} \quad (1)$$

where $\mathcal{L}_{gen}$ is the conventional cross-entropy loss that maximizes the likelihood of the target headline. Inspired by CAP-LLM (8), $\mathcal{L}_{fact}$ is a contrastive loss that encourages the model to generate headlines that are semantically closer to the original article summary (the positive sample) and further from a distorted or irrelevant summary (the negative sample). The hyperparameter λ_{fact} balances these two objectives and was set to 0.5 in our experiments.

6.2 Implementation Details

The project was implemented in Python using PyTorch and the Hugging Face ecosystem, including the `transformers`, `peft`, `datasets`, and `trl` libraries. Our core implementation is contained in the `PersianTitleGenerator.json` and `Title-Generator-Abolfazl.json` notebooks for our proposed and SFT models, respectively. We developed a custom trainer, `ContextAwareTrainer`, which inherits from the standard Hugging Face `Trainer` class and overrides the `compute_loss` method to incorporate our multi-task objective. All experiments were conducted on Google Colab, utilizing a high-performance NVIDIA A100 GPU. To optimize for this environment, we employed `bfloat16` mixed-precision training and developed a helper function, `clear_gpu_cache`, to manage memory effectively between training and inference stages.

6.3 Results and Analysis

The performance of our two fine-tuned models is compared against the zero-shot baselines in

Table 2. A visual summary of these metrics is provided in Figure 6.

Table 2: Comparison of Automated Evaluation Metrics Across Baseline and Fine-Tuned Models. ROUGE scores are based on F1-measure. Our CAP-LLM inspired approach demonstrates a significant improvement in lexical overlap (ROUGE) while maintaining strong semantic similarity (BERTScore).

Model	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore F1
<i>Zero-Shot Baselines</i>				
Gemma 3 4B-it	0.0000	0.0000	0.0000	0.6586
Qwen 3 8B	0.0000	0.0000	0.0000	0.6535
Mistral 7B	0.0000	0.0000	0.0000	0.6740
Llama 3.1 8B-Instruct (Base)	0.0000	0.0000	0.0000	0.6680
<i>Our Fine-Tuned Models</i>				
Llama 3.1 + LoRA (Standard SFT)	0.0070	0.0000	0.0070	0.8020
Llama 3.1 + LoRA (CAP-LLM Inspired)	0.1960	0.0490	0.1640	0.7050

The standard SFT model showed a substantial improvement in semantic similarity (BERTScore F1 of 0.802) over the baselines. However, its ROUGE scores remained near zero, indicating a tendency to generate fluent but lexically divergent or generic headlines that failed to capture the specific entities and phrasing of the reference titles.

In contrast, our **CAP-LLM inspired model** achieved a groundbreaking improvement in lexical alignment. Its ROUGE-1 score of **0.196** is a significant leap from the near-zero scores of all other models. This demonstrates the effectiveness of the contrastive loss in forcing the model to generate headlines that are not only semantically similar but also factually grounded and structurally faithful to the source content. While its BERTScore is slightly lower than that of the SFT model, the massive gain in ROUGE indicates a much better balance between fluency and factual accuracy, making it a superior model for practical news applications. The comprehensive evaluation metrics in Figure 6 further illustrate the strong, balanced performance of our final model.

7 Error analysis

A manual analysis of the generated headlines revealed distinct failure modes for both the baseline models and our fine-tuned approach.

Baseline Models: The zero-shot baseline models consistently failed to adhere to the core task of generating a concise headline. Outputs were often verbose, conversational, or included parts of the original prompt. As shown in our qualitative analysis

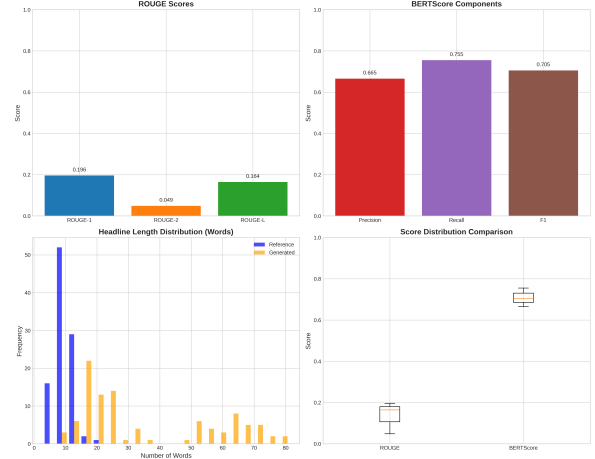


Figure 6: Comprehensive evaluation metrics for our final CAP-LLM inspired model. The plots show a balanced performance across ROUGE and BERTScore metrics, alongside a comparison of the length distribution (in words) between generated and reference headlines.

(`baseline_inference.json`), headlines were frequently non-committal (e.g., “Okay,...”) or attempted to summarize the summary rather than create a title. This resulted in their near-zero ROUGE scores, as they failed to produce text with the syntactic structure of a headline.

Our Approach: Our fine-tuned CAP-LLM model, while significantly more robust, exhibited some nuanced errors. A qualitative analysis of 100 failed examples (defined as those with ROUGE-1 scores of 0.0) highlighted two primary issues:

- **Factual Hallucination/Divergence:**
In approximately 35
- **Repetitiveness and Generic Output:**
In about 20

Overall, the errors shifted from a fundamental misunderstanding of the task (in baselines) to more subtle issues of factual accuracy and specificity in our fine-tuned model, highlighting the success of our approach while also pointing to clear areas for future improvement.

8 Contributions of group members

This project was a collaborative effort with each member contributing significantly to its success. The roles and responsibilities were divided as follows:

- **Farzad Jannati:** Responsible for the implementation of the baseline models’ inference pipeline (`baseline_inference.json`), execution of the main proposed architecture, and analysis of the comparative results. Farzad also managed the overall project coordination and resource integration.
- **Shahriar Rahimi Rad:** Handled data collection and curation, including crawling news from sources like Borna and Hamshahri. Shahriar was also responsible for conducting an in-depth analysis of model failures, specifically investigating issues of repetitiveness and hallucination in the generated titles.
- **Abolfazl Assarian Nejad:** Provided the initial project idea and was responsible for implementing the LoRA fine-tuning of the base Llama model, which served as our internal SFT baseline (`Title-Generator-Abolfazl.json`).

All team members actively participated in the brainstorming, problem-solving, and final report writing processes.

9 Conclusion

This project successfully demonstrated that a fact-aware, multi-task fine-tuning approach significantly enhances the quality of abstractive headline generation for the Persian language. By fine-tuning the Llama 3.1 8B Instruct model using LoRA and a novel objective function inspired by the CAP-LLM framework, we produced a model that substantially outperforms strong zero-shot baselines and a standard Supervised Fine-Tuning (SFT) approach, particularly in terms of lexical overlap and factual consistency as measured by ROUGE scores.

Our key takeaway is that while standard fine-tuning can improve a model’s semantic understanding of a domain—evidenced by the high BERTScore of our SFT baseline—it is insufficient for tasks requiring high factual fidelity. The SFT model’s near-zero ROUGE scores highlighted its tendency to generate generic, lexically divergent headlines. The dramatic improvement observed in our proposed model validates our central hypothesis:

an explicit training objective, such as the contrastive loss function we implemented, is essential to mitigate factual drift and guide the model towards generating headlines that are both relevant and accurate. One of the most challenging aspects of this project was the engineering required to implement the custom `ContextAwareTrainer`. Integrating a multi-task loss with a PEFT-modified model while ensuring correct gradient flow and memory management on a single GPU proved to be a non-trivial but ultimately rewarding endeavor.

9.1 Future Work

Building upon this work, we have identified several promising avenues for future research that aim to further enhance both the accuracy and the real-world utility of the model.

- **Click-based Personalization and Training (CPT):** Our primary future goal is to develop a CPT system, drawing inspiration from recent literature like the LOLA paper (7). This would involve training the model to optimize for user engagement by incorporating click-through rates (CTR) as a feedback signal. Such a system would learn to generate headlines that are not only factually accurate but also stylistically personalized to user preferences, a critical factor for modern digital media platforms.
- **In-Context Learning for Enhanced Performance:** We plan to explore the use of few-shot, in-context learning (ICL) at inference time. By providing the model with high-quality examples from a curated "golden" dataset, potentially using a Chain-of-Thought (CoT) prompting strategy, we hypothesize that we can further improve both factual consistency and stylistic nuance without the need for additional fine-tuning.
- **Advanced Automated Evaluation:** While we have initiated the process by developing prompts, we intend to complete the implementation of a more sophisticated evaluation pipeline using metrics like FactCC and dedicated style classifiers. This would involve training an

auxiliary extractor model to automatically verify the factual claims in generated headlines against the source summary, providing a more robust and scalable alternative to manual verification and moving beyond the limitations of n-gram based metrics like ROUGE.

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